

of the lot turned out to be Plextor. We were shocked to see its abysmal data transfer rate of 8X. Next in line was the Liteon 48X drive, with a poor transfer rate of 18.9X. The best of the lot was Asus, with a transfer rate of 39.04X. It was followed closely by the BenQ with 37.59X and the Liteon 52X, which scored 36.44X. Transfer times are most important for transferring data to a hard disk.

Next up were the access or seek time tests. Plextor disappointed us again, with an access time of 126 ms while BenQ came up tops with 82 ms. In real world performance, this translates into faster reading of a disc. The spin-down time was also noted.

Remember that for higher drive speeds, more spin-down time is required. Since all the drives had comparable speeds, it shouldn't have mattered here. Again we were in for a surprise—Asus was the worst with 10.56 seconds, BenQ was close behind with 6.82 seconds and Plextor made up for its former poor performances by topping this test with 0.01 seconds.

SiSoft Sandra was used to gauge the sequential and random read speeds. The sequential read speeds remained stagnant between 3.5 and 4 MBps for most drives. However, in the random read speeds, Krypton topped the tests with a

speed of 1,186 KBps. Samsung had the lowest score with 787 KBps, which was surprising considering its 8 MB buffer. Samsung also did badly in the access time test, with 254 ms. The honours for the fastest access time went to the Krypton 52X, followed by the Liteon 52X and the Plextor respectively.

Finally, we ran the extraction tests, in which the drives had to extract the first, middle and last track from a pressed audio CD. The AOpen drive was the fastest and the Asus drive came in last.

In order to test the drives for writing and re-writing, we did many tests involving the burning and ripping of audio and data CDs.

The first test was the sequential write test, in which we used a 700 MB movie file and timed the burning process. The Sony 52X proved to be the fastest, burning the file in 155 seconds, the Asus 52X was the slowest, taking 174 seconds.

The assorted CD burning test was a neck to neck competition. The LG 52X took the crown with 153 seconds, surpassing even its own sequential write time. Samsung timed a very poor 207 seconds, leaving us wondering where its 8 MB buffer went.

The CD-to-CD copy test proved to be the Liteon 52X's undoing, which took a



The Asus drive comes with a great bundle of goodies

Copy-Protection

Copy-protection is the technology devised to protect media content, be it streaming or static, from being copied. Although, it is okay to make one personal backup copy of any media for archival purposes, media companies are taking no chances and are aggressively using different copy protection techniques to protect their content. Some very popular techniques in use by game publishers are SafeDisc from Macrovision and SecuROM from Sony. Audio CDs use Cactus Data Shield by MidBar Technologies.

SafeDisc: SafeDisc 2 is the latest incarnation, and comprises three main features—an authenticating digital signature, encryption that protects the content and anti-hacking software.

SecuROM: SecuROM is a copy-

protection technique made by Sony and is employed by game publishers. The process involves adding a unique electronic SecuROM key to the CD-ROM using special glass-mastering equipment also manufactured by Sony DADC. The catch lies in the electronic key, which is difficult to copy to a CD-R.

These protection techniques can be worked around using certain copying software. CloneCD is one such one-to-one backup CD software that is commonly used. We tried copying a few copy protected discs using CloneCD, to get around Safedisc and Securom protection for all the drives we tested. All the drives were able to circumvent both Safedisc and Securom—except for the twins from BenQ, which failed both the tests, and the LG GCE-8523B which failed the Securom test.

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